IITM PRAVARTAK
CATALYSING INNOVATION

Professional Certificate Programme in Agentic AI and Applications

Introduction to Agentic AI

Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

- An overview of LangChain and how it enables LLM-powered applications
- How prompt formatting and response parsing shape reliable model interactions
- The principles behind effective prompt structuring for clarity and control
- How response parsing converts model outputs into usable, structured information
- How LangChain tools and tool-calling extend agent capabilities and workflows
- An introduction to prompt templates and how they standardise repeatable tasks

Topics Covered

- What is Agentic AI
- The Rise of Agentic AI
- The Agent Life Cycle and Capabilities
- Types of Autonomy and Maturity Levels
- Rule Based Systems v. Agentic AI
- Live Demo: From Goal to Action Plan

What is Agentic AI?

Agentic AI

Refers to systems that decide, plan, and act autonomously to achieve goals, not just answer questions

Instead of waiting for you to give every step, they:

- Understand a goal (e.g., “book me a flight”).
- Make a plan (search flights, compare options).
- Take action (book the ticket)

Agentic AI

Example

A normal AI chatbot only answers questions, while an agentic AI can take actions like ordering food, scheduling meetings or running code to solve tasks.

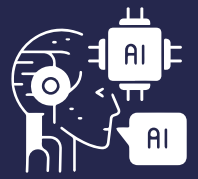
Agentic AI

Examples in Practice

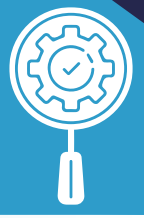
- **Software dev agent**: Reads bug reports, edits code, runs tests, pushes fixes
- **Business ops agent**: Monitors KPIs, generates reports, alerts managers
- **Personal AI assistant**: Books meetings, manages emails, orders supplies online

Understanding the Evolution

Agents v. Agentic AI



"Agents" and "agentic AI" sound alike but follow fundamentally different approaches to autonomy.



Recognise this distinction to design effective multi-agent systems.

Agents: The Foundation

- Perceive its environment, makes decisions and acts to achieve goals
- Range from simple rule-based systems to sophisticated programmes
- Operate autonomously but usually rely on pre-programmed responses and limited decision trees

Agents: The Foundation

Chess-playing bot

Evaluates positions and selects optimal moves using classical algorithms

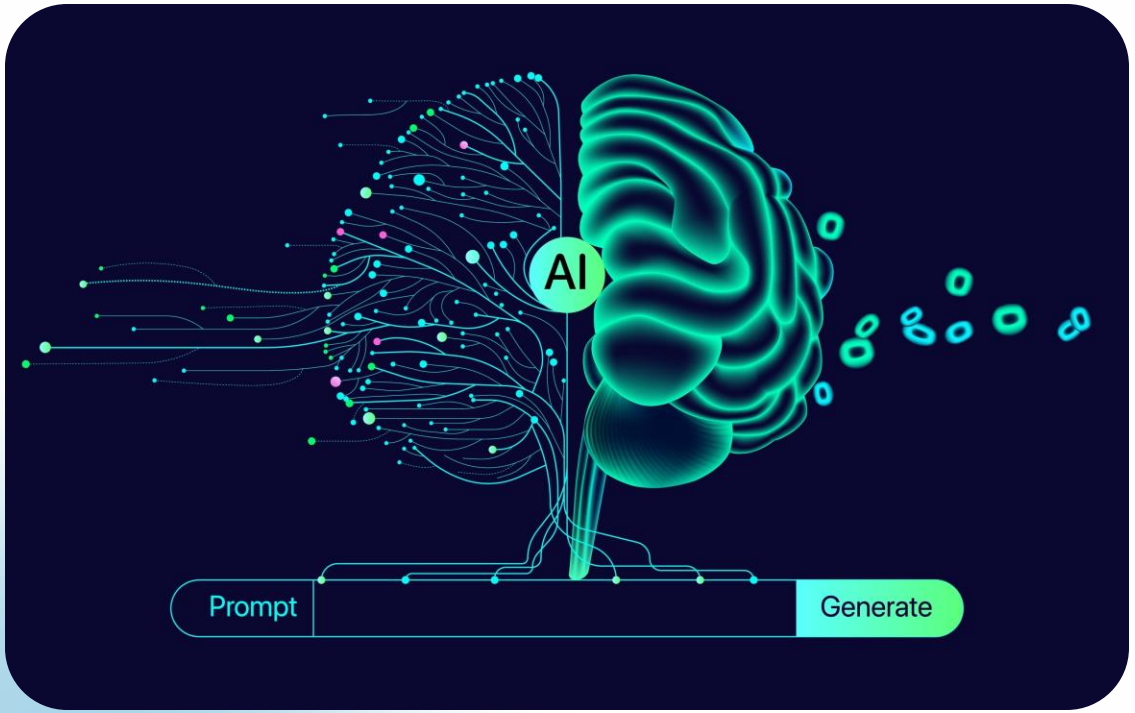
Roomba robot

Detects obstacles and follows pre-programmed cleaning patterns

Rule-based chatbot

Answers queries using ‘if-then’ logic trees and keyword matching

Agentic AI: The Next Evolution



- Refers to advanced agents powered by modern AI models like LLMs, reasoning engines and sophisticated planners
- Reasons, plans and adapts to achieve complex, multi-step goals

Agentic AI: The Next Evolution

Understand complex contexts and multi-modal inputs beyond basic sensors.

Develop advanced strategies and adapt plans to changing conditions.

Coordinate actions using multiple tools while continually improving.

Perceive & analyse

Reason & plan

Execute & learn

Real-World Applications: Seeing the Difference

Traditional agents

- FAQ chatbot with scripted responses
- Inventory tracker with threshold alerts
- Email autoresponder with templates
- Basic recommendation engine

Agentic AI systems

- Customer support AI queries CRM, escalates tickets and drafts personalised follow-ups
- Supply chain agent replans routes dynamically and coordinates with multiple vendors
- Research assistant searches, analyses, synthesises and presents findings

Key Differences: A Technical Comparison

Aspect	Traditional agents	Agentic AI
Core capability	Rule-based actions in narrow domains	LLM-powered reasoning with adaptive planning and tool use
Decision-making	Pre-programmed logic trees and fixed responses	Dynamic reasoning that adapts to context and goals

Key Differences: A Technical Comparison

Aspect	Traditional agents	Agentic AI
Task scope	Limited to specific, predefined functions	Open-ended, multi-step workflows with emergent capabilities
Learning	Minimal - primarily through manual updates	Continuous improvement through experience and feedback
Example analogy	Junior employee following a detailed checklist	Proactive manager who understands goals and adapts strategies

The shift is from reactive execution to proactive intelligence—agentic AI understands intent and pursues outcomes, not just instructions.

The Strategic Takeaway

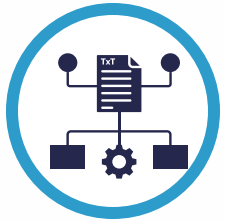
Traditional agents suit predictable tasks; agentic AI excels in complex, dynamic environments.

Agentic AI handles ambiguity, makes contextual decisions and coordinates workflows.

The shift from agents to agentic AI marks a move from automation to true AI teammates.

For system architecture

For team planning



For the future

The choice between agents and agentic AI shapes how autonomous systems fit into your organisation’s workflows and goals.

The Rise of Agentic AI

The Rise of Agentic AI

- Agentic AI fundamentally differs from traditional AI, marking a major evolutionary leap.
- It shifts AI from responding to commands to initiating actions and making autonomous decisions.

Agentic AI: The New Frontier in Artificial Intelligence

Definition: Autonomous AI systems that perceive, decide, act and learn independently without constant human supervision.

Market growth

- Market expanded from $1.4B in 2020 to $14.1B by 2025
- Represents a 58% compound annual growth rate

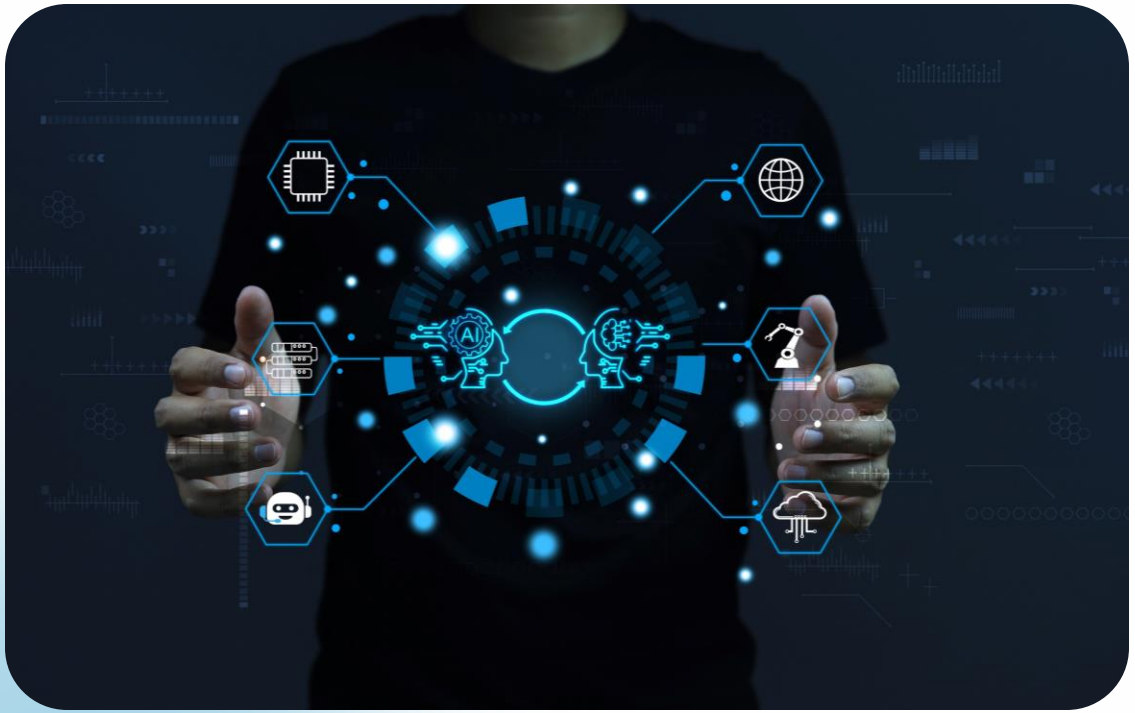
Core value

- Enables real-time adaptive decision-making beyond static programming
- Supports truly autonomous operation in complex environments

Single-Agent v. Multi-Agent Systems

Single-agent systems One AI entity handling all tasks independently	**Multi-agent systems** Collaborative agents coordinating workflows
• Unified decision-making process	• Specialised agents for different functions
• Simpler implementation and maintenance	• Enhanced resilience through redundancy
• Limited by individual agent capabilities	• Enables complex emergent behaviours
• Best for focused, specific applications	• Ideal for multifaceted problems

The Core Question: One Agent or Many?



- Understanding when to deploy single or multi-agent architectures is key to AI success.
- This framework guides engineering teams to align choices with project needs, resources and goals.

One Agent or Many

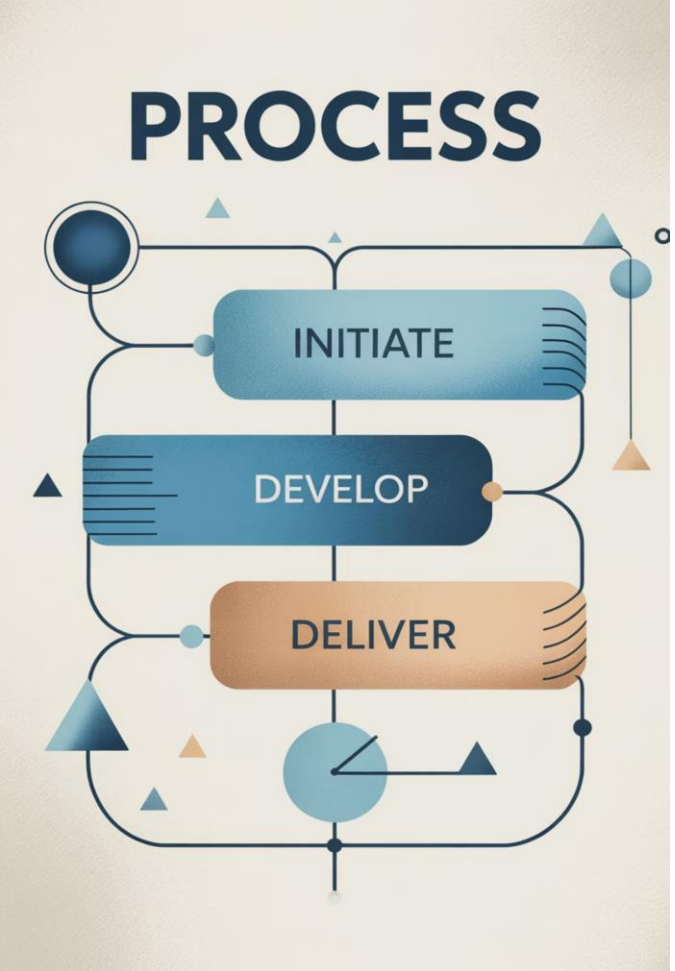
Single-Agent v. Multi-Agent Approach

Single-Agent approach
One intelligent agent completes the entire workflow, using reasoning and external tools to handle complex tasks.

Multi-Agent approach
Multiple specialised agents collaborate, each applying domain expertise to solve different parts of a complex problem.

Choosing between these architectures directly affects system complexity, scalability and performance.

Single-Agent Systems



- **Simple architecture**: Easy to design, deploy and monitor with minimal coordination
- **Sequential processing**: Ideal for linear workflows with predictable task order
- **Resource efficient**: Low computational cost, simple debugging and reduced maintenance overhead

Single-agent systems excel in well-defined domains that require no cross-disciplinary expertise, handling tasks effectively through a single reasoning entity.

Example of Single-Agent Systems

Customer Support

Key insight: For clear, sequential workflows without specialised expertise, single agents are optimal.

A single agent efficiently handles the entire support workflow:

- Receives and analyses customer inquiries
- Searches knowledge base for relevant information
- Drafts contextual responses
- Escalates complex issues when necessary

This works because each step builds on the last, making coordination unnecessary in a linear process.

Multi-Agent Systems: The Specialist Team

- **Divide & conquer**: Breaks complex problems into manageable parts, each handled by a specialised agent
- **Specialised accuracy**: Delivers precise, efficient results through focused domain expertise.
- **Natural scalability**: Expands effortlessly by distributing tasks across expert agents

This power brings coordination challenges, as agents must share information, avoid conflicts and align goals without falling into endless discussion loops.

Multi-Agent in Action: Drug Discovery Pipeline

Research agent: Analyses scientific literature to identify research directions and gaps

Chemistry agent: Designs novel compounds based on research, optimising for therapeutic properties

Simulation agent: Validates compound viability through toxicity and efficacy models before costly lab trials

Project manager agent: Integrates cross-domain results, monitors progress, and allocates resources strategically

No single agent can feasibly master chemistry, biology, simulation modelling and project management all at once.

Decision Framework: Choosing Your Architecture

Factor	Single-agent when	Multi-agent when
Task complexity	Task is narrow, well-defined, sequential	Task spans multiple domains or requires parallel subtasks
Resource constraints	Limited compute budget; need lightweight setup	Can afford orchestration overhead and need scale
Specialisation needs	One agent can reasonably handle expertise required	Tasks require diverse, deep domain expertise
Coordination requirements	Minimal; single reasoning chain sufficient	Multiple perspectives must be combined strategically

The Professional Implementation Strategy

Start single-agent

Begin with the simplest viable setup, faster to prototype, cheaper to run and easier to debug

Monitor complexity signals

Look for signs of strain: inconsistent outputs, reasoning failures or performance bottlenecks

Evolve to multi-agent

Shift when tasks exceed single-agent limits, demand parallelism or require modular expertise.

Remember: Architecture evolution is natural — successful AI systems often start simple and grow as requirements clarify.

Key Warning Signs for Architecture Change

Time to consider multi-agent

- Single agent struggles with domain expertise depth
- Tasks require simultaneous parallel processing
- Quality drops on long reasoning chains
- Different subtasks need conflicting optimisation approaches

Stick with single-agent

- Workflow remains fundamentally sequential
- Domain knowledge requirements are manageable
- Coordination costs outweigh the benefits
- Budget or timeline limits favour a simpler approach

Your Architecture Decision Playbook

Swiss Army Knife Rule

Single agents are like Swiss Army knives - versatile enough for most everyday jobs, perfect when simplicity and reliability matter most.

Specialist Team Rule

Multi-agent systems are like assembling a team of specialists - necessary for big, multi-dimensional problems that demand deep expertise.

Next Steps for Your Team

Evaluate current use cases against the framework

Begin with single-agent prototypes for fast, low-cost validation

Scale to multi-agent only when complexity demands it

Remember: the best architecture is the simplest one that meets your requirements.

Why Agentic AI Matters Today

Cost reduction (30%)

Achieves up to 30% lower operational costs through intelligent automation

Continuous operation (24/7)

Handles complex workflows end-to-end, enabling 24/7 productivity without constant human input

Decision quality (3X)

Improves decision-making up to 3× through data-driven insights and scenario analysis

Agentic AI is transforming healthcare, manufacturing automation, and corporate learning through targeted, domain-specific impact.

The Agent Life Cycle & Capabilities

The Agent Life Cycle & Capabilities

- Understand the core cycle driving agentic AI systems
- Explore the capabilities of single-agent and multi-agent architectures

The Perceive → Decide → Act → Learn Cycle

Perceive

- Gather data from sensors, databases, user inputs or digital environments
- Process and filter information for relevance and priority

Decide

- Analyse current state against goals and constraints
- Plan next steps based on available options and predicted outcomes

Act

- Execute tasks or interventions autonomously
- Implement decisions through digital or physical means

Learn

- Evaluate results against expectations
- Update knowledge base and improve future decisions

This cycle allows agentic AI to self-improve and adapt to changing environments.

Autonomy and Adaptability in Action

Agentic AI stands out by continuously monitoring its environment and using feedback loops to guide decisions.

Unlike traditional systems, agentic AI can independently:

- Detect environmental changes affecting performance
- Evaluate current strategy effectiveness
- Adjust approaches without human reprogramming
- Optimise for short-term outcomes and long-term goals

What Does Autonomy Mean

Autonomy is the ability to act independently without step-by-step instructions

In AI, it reflects how freely a system chooses actions to meet objectives

The Core of Autonomous Systems

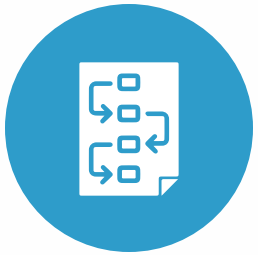
Assess options and select the best path based on conditions and goals

Outline steps to reach the outcome, factoring in obstacles and resources

Carry out the plan, adapt in real-time and learn from each experience

Decide

Plan



Act

A system is autonomous if, once instructed, it independently performs tasks and determines next steps based on its environment and goals.

Understanding Through Everyday Examples

Autonomous v. Non-Autonomous Systems

Autonomous systems	Non-autonomous systems
Interpret high-level goals and adapt dynamically by deciding, planning, and acting based on environmental conditions.	Follow step-by-step instructions and cannot adjust beyond predefined programming.

Understanding Through Everyday Examples

Non-autonomous

Microwave oven

- Requires manual input for time and power level
- Executes fixed instructions without deviation
- Lacks awareness of the food's actual cooking state

Autonomous

Self-driving car

- Accepts high-level goals like destination address
- Interprets sensor data to plan and adjust actions
- Navigates and reacts autonomously to real-world events

Static Predictions vs. Dynamic Decision-Making

Traditional AI	Agentic AI
Executes predefined algorithms and patterns	Plans and strategises proactively to meet goals
Reacts passively within programmed limits	Evaluates options and chooses optimal paths
Relies on static predictions from past data	Adapts actions in real-time to changing conditions

Real world example: Traditional AI categorises support tickets; agentic AI diagnoses issues, acts autonomously and learns over time.

Types of Autonomy and Maturity Levels

Types of Autonomy and Maturity Levels

- Explore the spectrum of autonomy in agentic AI systems
- Distinguish between reactive agents and autonomous decision-makers
- Assess and enhance your organisation’s AI maturity

Four Levels of AI Autonomy

Reactive	Assisted	Semi-autonomous	Fully autonomous
• Responds instantly to stimuli without memory or planning • Example: Thermostat switching heating on/off by temperature	• Provides suggestions in human-in-the-loop with AI systems • Example: Driver-assist alerts without vehicle control	• AI handles tasks with occasional human intervention • Example: Drones navigating independently but needing approval for key decisions	• AI operates independently in complex, dynamic environments • Example: Self-driving cars navigating urban traffic without human input'

Maturity Spectrum: From Rule-Based to Fully Agentic

Early AI

- Rule-based systems with scripted responses and limited flexibility
- Operates within narrow, predefined parameters

Emerging autonomy

- Systems with limited learning capabilities that adapt within specific domains
- Requires regular human supervision and guidance

Advanced AI

- Self-improving systems with robust learning mechanisms
- Handles complex scenarios with minimal human intervention

Fully agentic

- Goal-driven, adaptive systems capable of long-term planning
- Continuously self-optimises performance across varied environments

Rule-Based Systems v. Agentic AI

Rule-Based Systems v. Agentic AI

- Compare traditional rule-based systems with modern agentic AI.
- Recognise the limitations of fixed programming approaches.
- Understand the transformative potential of adaptive, autonomous systems.

Rule-Based Systems

Operating Principles

- Functions through fixed "`if-then`" rules and predefined decision trees
- Cannot operate outside explicitly programmed parameters

Rule-Based Systems

Example Scenario

Robot vacuum programmed with simple rules:

- If bump detected, turn right
- If battery below 20%, return to dock
- If scheduled time, begin cleaning

Rule-Based Systems

Key Limitations

- Fails to adapt to new or unexpected situations
- Depends on manual updates to change behaviour
- Operates only within predefined scenarios
- Lacks capacity to learn or optimise over time

Agentic AI: Adaptive and Proactive

Operating Principles

- Learns from environment and experience to continuously improve
- Plans multi-step actions toward defined goals
- Adapts strategies based on feedback and changing conditions

Agentic AI: Adaptive and Proactive

Example Scenario

AI-powered vacuum with agentic capabilities:

- Maps room layout to plan optimal cleaning path
- Recognises furniture types and adjusts cleaning approach
- Learns from cleaning history to prioritise high-traffic areas
- Identifies and remembers recurring obstacles

Agentic AI: Adaptive and Proactive

Key Benefits

- Handles novel situations with flexibility
- Optimises efficiency through continuous improvement
- Minimises reliance on human input or programming
- Enhances outcomes through experiential learning

Rule-Based v. Agentic AI in Action

Rule based approach	Agentic approach
Fixed cleaning pattern regardless of room layout	Dynamically generated cleaning path optimised for efficiency
Repeats same mistakes across multiple cleaning sessions	Improves performance with each cleaning cycle
Treats all obstacles identically	Differentiates between obstacle types and adjusts behavior
Requires manual intervention when stuck	Self-recovers from most challenging situations
Uniform cleaning time allocation regardless of need	Prioritises areas based on usage patterns and dirt detection

Agentic AI turns static automation into intelligent autonomy through learning and strategic planning.

The Future is Agentic AI

Gartner predicts one-third of applications will have agentic AI by 2028

By automating 15% of tasks, agentic AI frees knowledge workers for higher-value work

Automated experimentation with agentic AI accelerates innovation by 40% in organisations

Software integration (33%)

Task automation (15%)

Innovation acceleration (40%)

As agentic systems evolve, they will drive smarter, faster and more resilient organisations across all sectors.

Conclusion: Embrace the Agentic AI Revolution

Paradigm shift
Agentic AI marks a fundamental shift from automation to adaptive, value-driven intelligence

Competitive advantage
Adopting agentic AI unlocks agility, cost efficiency, and lasting competitive edge

Call to action
Now is the time to explore tools and invest in autonomous, AI-led transformation

Key Takeaways: The Agentic AI Revolution

What we covered:

- Definition and importance of agentic AI in today's landscape
- Fundamental differences between traditional and autonomous AI
- The perceive-decide-act-learn cycle that powers agentic systems
- Progression from reactive to fully autonomous AI capabilities
- How agentic AI transforms rule-based approaches through adaptive learning

What this means for you:

- Opportunity to automate increasingly complex workflows
- Potential for significant cost reduction and efficiency gains
- Competitive advantage through early adoption
- Need for strategic planning around AI implementation
- Importance of understanding autonomy levels for responsible deployment

Getting Started: Your Agentic AI Journey

Assess current AI maturity

- Evaluate existing implementations for agentic opportunities
- Identify limitations caused by rule-based systems

Identify pilot projects

- Select contained use cases with measurable success criteria
- Prioritise areas where adaptability adds immediate value

Build capabilities

- Develop internal expertise or engage specialist partners
- Establish governance for responsible autonomous deployment

Scale and optimize

- Extend successful use cases across organisational functions
- Continuously monitor performance and refine strategy

Limitations & Risks of Agentic Systems

Hallucinations & reliability: Prone to generating incorrect or fabricated outputs

Safety & guardrails: Risk of harmful, biased, or unsafe actions without robust controls

Coordination challenges: Agents may conflict, duplicate work or mismanage dependencies

Over-autonomy risks: Unintended decisions or escalation without human oversight

Data & privacy concerns: Exposure or misuse of sensitive information during operation

Transforming Operations & Supply Chain

Inventory management agents

These intelligent systems streamline stock control by combining real-time monitoring, predictive analytics and automated actions.

- Real-time stock monitoring across multiple locations
- Predictive demand forecasting using historical and market data
- Automated reorder triggers to prevent stockouts and reduce excess

Logistics optimisation agents

These adaptive systems enhance delivery efficiency by responding to real-time conditions and reallocating resources dynamically.

- Dynamic route optimisation based on traffic, weather, and constraints
- Real-time adaptation to delays and disruptions
- Resource allocation optimisation for maximum efficiency

Result: Organisations achieve a 15–30% reduction in operational costs while improving delivery reliability and customer satisfaction.

Financial Intelligence & Compliance Automation

Financial planning agents

40%

Risk reduction

Faster anomaly detection

- Analyse transaction patterns and detect spending anomalies
- Provide cost-saving insights and suggest strategic decisions
- Identify fraud in real time based on market conditions

Regulatory compliance agents

60%

Time savings

Automated reporting

- Monitor evolving regulatory landscapes continuously
- Update compliance protocols and generate reports automatically
- Reduce compliance costs while ensuring regulatory alignment

HR & Workforce Optimisation

- Screens thousands of CVs using advanced matching algorithms
- Identifies top candidates based on skills, experience and cultural fit
- Automates interview scheduling and delivers detailed assessments

- Analyses performance data to identify skill gaps
- Recommends tailored training programmes for each employee
- Adapts learning paths based on progress and evolving business needs

- Tracks engagement, productivity, and career progression
- Provides actionable insights for team development and retention
- Supports strategic workforce planning through data-driven analysis

Intelligent recruitment

Personalised learning paths

Performance analytics

Research & Knowledge Management Revolution

Market research agents

- Monitor competitor activities and analyse market trends from multiple data sources
- Provide real-time competitive intelligence and identify emerging opportunities early

Rag knowledge agents

- Retrieve relevant information from large document repositories
- Generate contextual insights to make institutional knowledge accessible

"Knowledge agents democratise expertise, enabling faster decision-making and reducing dependency on subject matter experts for routine inquiries."

Live Demo: Task Planner Agent

From Goal to Action Plan

Let's see how an intelligent agent can break down complex goals into actionable subtasks, demonstrating the power of dynamic planning over static responses.

Demo Setup: Building Our Task Planner

- **Technical stack**
  - Langchain framework for agent orchestration
  - Openai gpt-4o-mini for reasoning
  - Custom task management tool
  - Zero-shot react agent pattern
- **Core components**
  - Task storage system (simple list)
  - Agent reasoning engine
  - Tool calling interface
  - Execution trace logging

PL will show a Demo

Expected Agent Behaviour & Output

Goal analysis
Agent interprets "team meeting for Friday" and identifies all necessary preparation steps

Task generation
Breaks down into specific, actionable subtasks with logical sequencing

Tool execution
Calls Task Manager tool multiple times to add each identified subtask

Key Demo Insights

Dynamic intelligence

- Demonstrates true reasoning beyond scripted responses
- Analyses goals, considers context, and generates tailored action sequences

Tool integration

- Seamlessly connects with external tools like databases, APIs and schedulers
- Scales to complex systems, enabling real actions in business environments

Adaptable planning

- Adjusts plans dynamically when goals change mid-demo
- Handles varied, real-world scenarios with flexible execution

Your Turn: Hands-On Agent Experience

Interactive challenge

- Provide a personal goal for the agent to interpret
- Watch it break down the goal into actionable subtasks
- Experience the practical value and adaptability of intelligent agents

Examples to use:

- “Organise a study group for our next exam“
- "Plan a weekend team-building event“
- "Set up a new development environment“
- "Launch a small product prototype"

Q & A

Thank you